Part I — GRPO Finetuning of `gemma-3-1b-it` on GSM8K

Multi-Agent Systems and Agentic AI — Practical Examination

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Jointly owned (shared team work): the v6e-1 TPU, the six shared GRPO runs R0–R5, and the code repository. All prose, figure selection and interpretation below are my own.

Code: `github.com/wdk0082/a8-grpo` *(to be ported to GitLab for submission)* | W&B: `a8-grpo` | GRPO, DAPO, LoRA, GSM8K

I.1 Reproducing the baseline

(i) The run. We reproduce the baseline as run **R0** (`8c2785ut`): upstream commit `borisbolliet/tpu-2026@77c5a67` with the `config.py` defaults (β=0.08, G=2), full **3364** GRPO steps (one epoch), $\sim$**5 h 05 m** wall-clock on `v6e-1`.

(ii) Accuracy. Held-out GSM8K *test*, HF split, fixed seed 42, greedy/exact-match. We report both the `config.py` default n=64 and the full n=1319 (for completeness; I.3 standardises on the full test) as *(n=64 / n=1319)*: base **45.3 / 47.4%**; R0 best-val **42.2 / 46.7%** (step 700) and final **0.0 / 6.2%** (step 3364). At either n the baseline *over-optimises* — R0 stays within noise of base then collapses by the final step (the format reward saturates while correctness decays); the wide n=64 CIs ($\pm$12 pt) are why I.3 uses the full test (per-checkpoint CIs in Table 1).

(iii) Training curves. Fig. 1 (R0, black): mean reward $\bar{r}$ rises, peaks $\approx$ step 1000, then declines to $\sim$0.5; $\mathrm{KL}(\pi_\theta \| \pi_{\mathrm{ref}})$ climbs from 0 (an early spike to $\sim$4, settling $\sim$1.6) — the over-optimisation arc.

Baseline patches (as-shipped fixes; full diff: `tpu-2026_our_changes.patch` in repo). (1) `run_tmux.sh` hard-coded a foreign home path; (2) the W&B entity defaulted to a third party — redirected to `a8-grpo`; (3) `evaluate.py` built a fresh B=0 LoRA and never restored a checkpoint, so it could only score *base* — we added Orbax `-step`/`-ckpt-dir` LoRA restore; (4) TFDS's gsm8k builder crashes under protobuf 7.x once JAX imports, so we evaluate on the identical GSM8K test via an HF `datasets` fallback (seed 42). (5) Raised checkpoint retention (`SAVE_INTERVAL_STEPS` 500$\to$100, `MAX_TO_KEEP` 4$\to$40) so the best-val checkpoint survives for selection — a persistence change, not a training one.

I.2 Understanding the baseline

(a) Policies & trainable parameters. π_θ is the LoRA-adapted policy: frozen `gemma-3-1b-it` weights plus low-rank adapters $\Delta W = BA$; **only** A, B **are trainable** (the base is frozen). π_{old} is the policy that generated the current rollouts — with `NUM_ITERATIONS` μ=1 there is one gradient step per rollout batch, so $\pi_{\mathrm{old}} = \pi_\theta$ at the update. π_{ref} is the frozen initial policy for the KL penalty; LoRA initialises B=0, so $\pi_\theta = \pi_{\mathrm{ref}}$ exactly at step 0.

(b) Group size & estimator. K=`NUM_GENERATIONS`= 2. Tunix uses `advantage_estimator='grpo'`: the **standard** group-mean, std-normalised advantage $\hat{A}_i = (r_i - \bar{r})/\sigma_r$ with $\bar{r}, \sigma_r$ over all K samples *including* i — *not* leave-one-out.

(c) Shaping vs. correctness rewards. Shaping: `match_format_exactly` and `match_format_approximately` (reward the <reasoning>/<answer> template). True correctness: `check_answer` and `check_numbers` (extracted number vs. gold). With the correctness reward *alone*, an early policy is almost never numerically right, so every sample in a group scores $\approx$ 0: within-group $\sigma_r \to 0$ and $\hat{A}_i = (r_i - \bar{r})/\sigma_r$ has no signal (an ε in the denominator averts a literal 0/0) — degenerate groups, no gradient. The shaping terms give differently-formatted samples different rewards, restoring $\sigma_r > 0$ and a usable advantage that bootstraps learning (the same $\sigma_r \to 0$ degeneracy analysed in I.4 Q1).

(d) Clipped surrogate & ϵ. The PPO-style clip lives in Tunix's `GRPOLearner` (module `tunix.rl`), in its clipped policy-gradient loss: it forms a **per-token** ratio $\rho_t = \pi_\theta(a_t)/\pi_{\mathrm{old}}(a_t)$ and optimises $\min\big(\rho_t \hat{A},\ \mathrm{clip}(\rho_t, 1-\epsilon, 1+\epsilon)\hat{A}\big)$, with ϵ=`EPSILON`= 0.2. Two deviations from the lecture form: (i) the ratio is **per-token**, not per-sequence; (ii) at μ=1, $\pi_{\mathrm{old}} = \pi_\theta \Rightarrow \rho_t \equiv 1$, so the clip is **dormant** in the baseline (it engages only for $\mu > 1$).

Table 1: Held-out accuracy on the *full* GSM8K test (n=1319, HF split, seed 42), greedy/exact-match, ±95% bootstrap CI. Last column: *paired* bootstrap Δ vs. base over the shared prompts (`paired_ci.py`). $*$ = paired 95% CI excludes 0; † = collapse. R1($G2, \beta.3$) and R2($G2, +\text{len}$) fell *below* base (R1 best 37.2 / final 0.0; R2 best 32.8 / final 40.6) — clean negative results.

Model	(G, β), step	Acc. (%)	95% CI	Δ base (paired)
Base `gemma-3-1b-it`	—	47.4	44.7–50.0	—
R0 (baseline)	$(2, .08)$ best 700	46.7	44.0–49.4	−0.7 n.s.
R0	$(2, .08)$ final 3364	6.2	4.9–7.5	$-41.2^{\dagger}$
R4	$(2, 0)$ best 1300	52.8	50.1–55.5	$+5.4^{*}$
R4	$(2, 0)$ final 3364	48.6	45.9–51.3	+1.2 n.s.
R3b	$(8, 0)$ best 1000	55.7	53.0–58.2	$+8.3^{*}$
R3b	$(8, 0)$ **final 3364**	**57.2**	54.6–59.9	$+9.9^{*}$
R5	$(8, .08)$ best 1300	53.4	50.7–56.2	$+6.1^{*}$
R5	$(8, .08)$ final 3364	55.6	52.8–58.2	$+8.2^{*}$

I.3 Improving on the baseline

Modification & motivation. The baseline over-optimises at K=2. The theory of I.4 Q1 gives the GRPO gradient variance $\mathrm{Var}(\hat{g}) = K^{-2} \sum_i (h_i - \bar{h})(h_i - \bar{h})^\top = O(1/K)$, with effective sample size $K_{\text{eff}} \propto K$: a larger group directly lowers the variance of the gradient estimate $\hat{g}$. We therefore take **group size** G (the K of I.2(b) and the theory) as the primary lever and run a controlled $\beta \times G$ **2×2** (KL-penalty weight × group size) to separate it from the KL leash: R4($G2, \beta0$), R0($G2, \beta.08$; the baseline), R3b($G8, \beta0$), R5($G8, \beta.08$) — plus single-axis controls R1(β=0.3) and R2(+soft length penalty), for **six full runs**. All share data (HF GSM8K, seed 42), eval, horizon (3364 steps) and a seed-matched rollout RNG (`PRNGKey(0)`); only the named knob changes per run.

We report **(a)** held-out accuracy (Table 1), **(b)** mean reward and KL for all runs on shared axes (Fig. 1), and **(c)** a diagnostic for a known GRPO failure mode — group degeneracy at G=8 (Fig. 2b).

(d) Findings vs. theory. *G=8 is the winning intervention.* Both G=8 runs beat base — R3b +9.9, R5 +8.2 pp, paired CIs excluding 0 — and each beats its matched G=2 run (R3b−R4 = +8.6 pp at final; R5−R0 = +49 pp). **R3b (G=8, β=0) is the best model at 57.2%.** The 2×2 (Fig. 2a) localises the failure: *the collapse is confined to the G=2, β>0 corner* (R0 final 6.2%, R1 0%), while the *same* β=0.08 at G=8 (R5) is stable, and at G=8 the KL weight is statistically irrelevant (R3b vs. R5: +1.7 pp, n.s.). Furthermore *over-optimisation is itself a G=2 phenomenon*: G=2 peaks at best-val then degrades (R4 52.8→48.6; R0 collapses), whereas G=8 **improves monotonically to the final step** (final > best). This is precisely the I.4 Q1 mechanism: larger K shrinks $\mathrm{Var}(\hat{g})$, so updates are smoother — the policy neither reward-hacks into R0's format-only collapse nor (once β>0) trips the exp(log-ratio) KL-estimator instability behind R1's blow-up.

Contradicted prediction. We predicted an over-optimising baseline would need a *tighter* KL leash. The opposite held: raising β to 0.3 (R1) collapsed *faster* (0%), and removing the KL term entirely (R4, β=0) was fine — the cure was variance reduction (group size), not regularisation, echoing DAPO's removal of the KL term for long-output RL. The diagnostic (Fig. 2b) shows the cost of G=8: $\approx \frac{1}{3}$ of groups are degenerate (all-K-equal ⇒ zero advantage), the K_{eff}/dead-group issue of I.4 Q1 and the target of DAPO's Dynamic Sampling — yet the net effect of more rollouts is still +8–10 pp. R1 and R2 fell below base (Table 1), honest negative results that further support variance, not extra shaping or a stronger leash, as the operative lever.

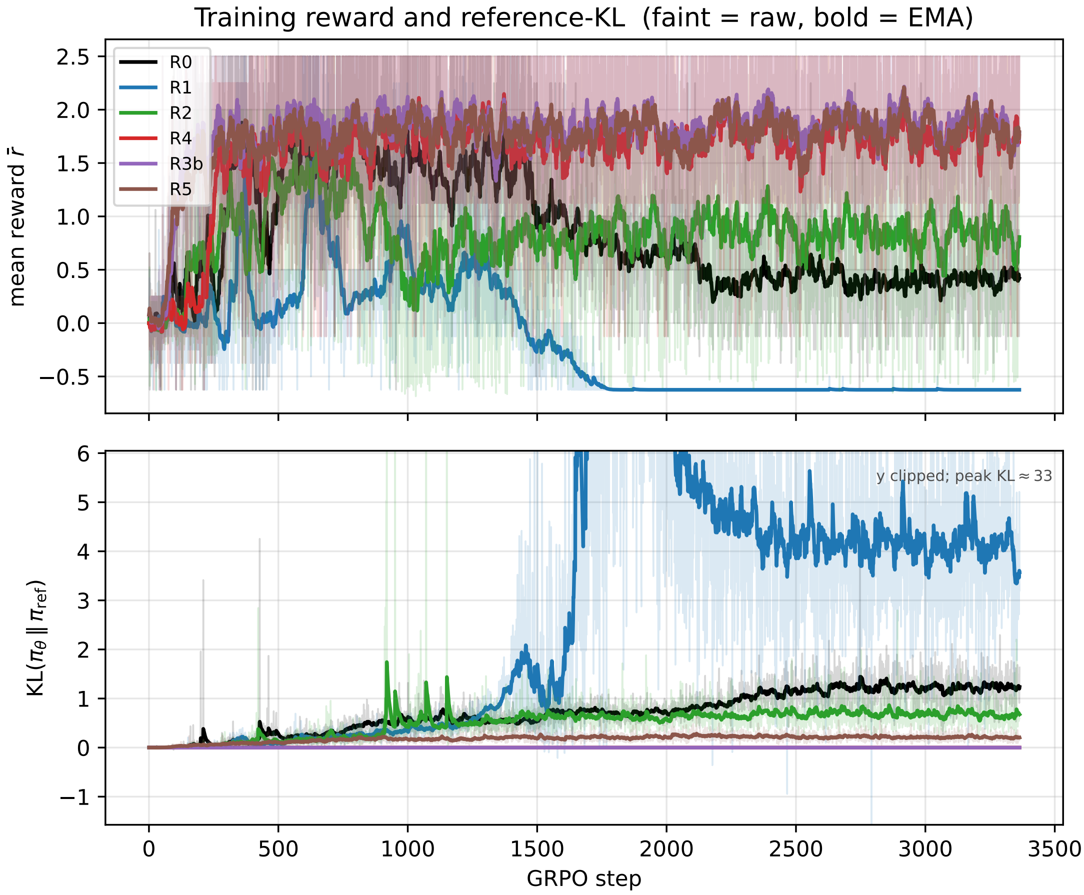
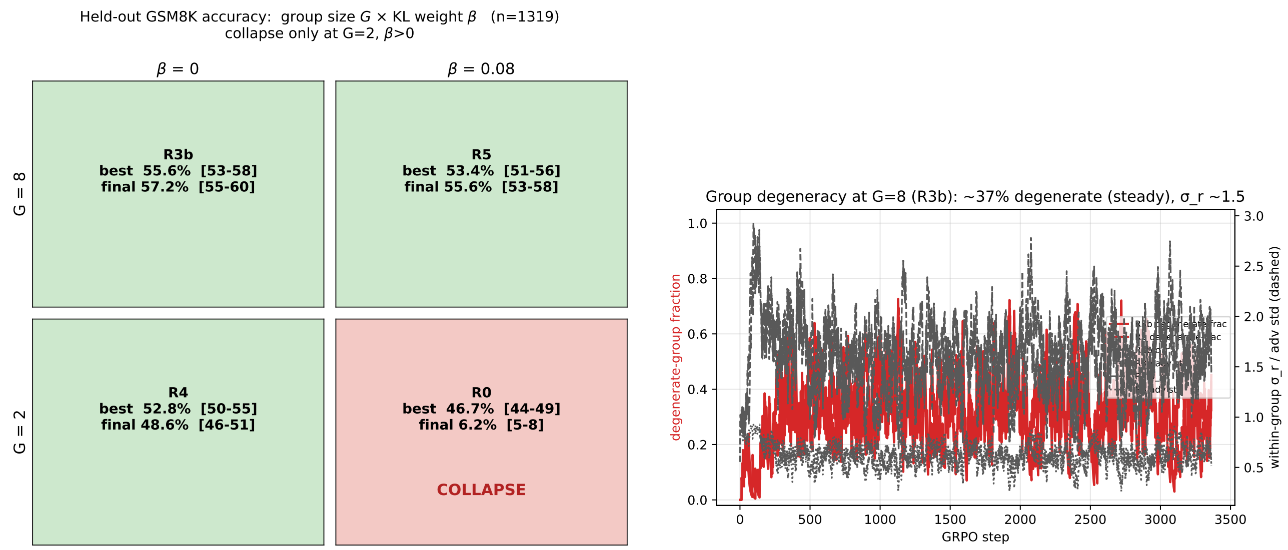


Figure 1: Mean training reward $\bar{r}$ (top) and $\mathrm{KL}(\pi_\theta \| \pi_{\mathrm{ref}})$ (bottom) vs. GRPO step, for all six runs on shared axes. R0 over-optimises (reward peaks then declines) and R1 collapses (KL blow-up $\sim$step 1700); the G=8 runs (R3b, R5) and R4 stay stable.



(a) $\beta \times G$ 2×2 (best/final acc).

(b) Group degeneracy at G=8.

Figure 2: *Left:* accuracy by (G, β) — collapse occurs *only* at G=2, β>0 (R0). *Right:* the diagnostic — at G=8 about a third of groups are degenerate (all-K rewards equal) yet σ_r holds ≈ 1.5 (steady, not $\to 0$).

I.4 GRPO theory — written questions (40 marks)

I.4.1 Group-mean normalisation

Let

$$\bar{h} = \frac{1}{K}\sum_{i=1}^{K} h_i.$$

(i) Since $\mathbb{E}[r_i] = \mathbb{E}[\bar{r}] = \mu$,

$$\mathbb{E}[\hat{A}_i] = \frac{\mathbb{E}[r_i - \bar{r}]}{\sigma} = 0.$$

Because $X_i = h_i \hat{A}_i$ and h_i is fixed, linearity of expectation gives

$$\mathbb{E}[X_i] = h_i\,\mathbb{E}[\hat{A}_i] = h_i \cdot 0 = 0, \qquad \boxed{\mathbb{E}[X_i] = 0}.$$

Summing over the group and again using linearity,

$$\mathbb{E}[\hat{g}] = \mathbb{E}\left[\frac{1}{K}\sum_{i=1}^{K} X_i\right] = \frac{1}{K}\sum_{i=1}^{K}\mathbb{E}[X_i] = \frac{1}{K}\sum_{i=1}^{K} 0 = 0, \qquad \boxed{\mathbb{E}[\hat{g}] = 0}.$$

Every expectation in this question is taken conditional on the sampled responses $a_1, \ldots, a_K$, which are held fixed; that is, $\mathbb{E}[\,\cdot\,]$ abbreviates $\mathbb{E}[\,\cdot \mid a_1, \ldots, a_K]$, so each score h_i is constant and only the artificial reward noise is random. In the usual policy-gradient problem the responses are sampled from the policy and their rewards depend on the sampled responses, so the policy gradient need not vanish.

(ii) The common random term is the sample mean $\bar{r}$, appearing in every

$$X_i = \frac{h_i}{\sigma}(r_i - \bar{r}).$$

For $i \neq j$,

$$\begin{aligned}
\mathrm{Cov}(r_i - \bar{r}, r_j - \bar{r}) &= \mathrm{Cov}(r_i, r_j) - \mathrm{Cov}(r_i, \bar{r}) - \mathrm{Cov}(\bar{r}, r_j) + \mathrm{Var}(\bar{r}) \\
&= 0 - \frac{\sigma^2}{K} - \frac{\sigma^2}{K} + \frac{\sigma^2}{K} \\
&= -\frac{\sigma^2}{K}.
\end{aligned}$$

Hence

$$\boxed{\mathrm{Cov}(X_i, X_j) = -\frac{1}{K} h_i h_j^\top, \qquad i \neq j.}$$

More precisely, the scalar covariance of the centred rewards is negative; the cross-covariance matrix is a negative scalar multiple of $h_i h_j^\top$. The negative dependence follows from the exact constraint

$$\sum_{i=1}^{K} (r_i - \bar{r}) = 0:$$

if one centred reward increases, the others must collectively decrease.

(iii) The full covariance matrix is

$$\boxed{\mathrm{Cov}(\hat{g}) = \frac{1}{K^2}\left[\sum_{i=1}^{K} h_i h_i^\top - \frac{1}{K}\left(\sum_{i=1}^{K} h_i\right)\left(\sum_{j=1}^{K} h_j\right)^\top\right] = \frac{1}{K^2}\sum_{i=1}^{K}(h_i - \bar{h})(h_i - \bar{h})^\top.}$$

For the scalar variance convention used in the lectures, define

$$\mathrm{Var}_{\mathrm{tr}}(Z) := \mathrm{tr}(\mathrm{Cov}(Z)) .$$

Taking the trace gives

$$\boxed{\mathrm{Var}_{\mathrm{tr}}(\hat{g}) = \frac{1}{K^2}\sum_{i=1}^{K} \|h_i - \bar{h}\|^2.}$$

Also,

$$\mathrm{Cov}(X_1) = \left(1 - \frac{1}{K}\right) h_1 h_1^{\top},$$

and hence

$$\mathrm{Var}_{\mathrm{tr}}(X_1) = \left(1 - \frac{1}{K}\right) \|h_1\|^2.$$

If one instead averaged K independent copies of X_1, the naive i.i.d. variance would be

$$\mathrm{Var}_{\mathrm{tr,iid}}(\hat{g}) = \frac{1}{K}\,\mathrm{Var}_{\mathrm{tr}}(X_1).$$

Following the lecture convention, define the effective sample size by

$$\mathrm{Var}_{\mathrm{tr}}(\hat{g}) = \frac{\mathrm{Var}_{\mathrm{tr}}(X_1)}{K_{\mathrm{eff}}}.$$

Therefore

$$\boxed{K_{\mathrm{eff}} = \frac{\mathrm{Var}_{\mathrm{tr}}(X_1)}{\mathrm{Var}_{\mathrm{tr}}(\hat{g})} = \frac{K^2\left(1 - \frac{1}{K}\right)\|h_1\|^2}{\sum_{i=1}^{K}\|h_i - \bar{h}\|^2} = \frac{K(K-1)\|h_1\|^2}{\sum_{i=1}^{K}\|h_i - \bar{h}\|^2}.}$$

Equivalently, its discrepancy from the nominal sample size K is

$$\frac{K_{\mathrm{eff}}}{K} = \frac{\mathrm{Var}_{\mathrm{tr,iid}}(\hat{g})}{\mathrm{Var}_{\mathrm{tr}}(\hat{g})}.$$

If

$$\frac{1}{K}\sum_{i=1}^{K}\|h_i - \bar{h}\|^2$$

remains of constant order and $\|h_1\|^2 = O(1)$, then

$$K_{\mathrm{eff}} = \Theta(K)$$

as $K \to \infty$.

If all score vectors lie in one direction, write

$$h_i = c_i v.$$

Then

$$\mathrm{Cov}(\hat{g}) = \frac{vv^{\top}}{K^2}\sum_{i=1}^{K}(c_i - \bar{c})^2,$$

so the covariance has rank at most one, and

$$K_{\mathrm{eff}} = \frac{K(K-1)c_1^2}{\sum_{i=1}^{K}(c_i - \bar{c})^2}.$$

Thus only variation in the scalar coefficients c_i contributes to the variance. In the special case $h_i = h$ for every i,

$$\hat{g} = \frac{h}{K}\sum_{i=1}^{K}\hat{A}_i = 0$$

for every reward realisation. Hence $\mathrm{Cov}(\hat{g}) = 0$ and, provided $h \neq 0$, the effective sample size is formally infinite.

I.4.2 KL-regularised policy update

(i) The objective is

$$F(\pi) = \sum_a \pi(a) A_t(a) - \frac{1}{\eta} \sum_a \pi(a) \log \frac{\pi(a)}{\pi_t(a)}.$$

Introducing a Lagrange multiplier λ for $\sum_a \pi(a) = 1$, stationarity on the support of π_t gives

$$A_t(a) - \frac{1}{\eta} \left(\log \frac{\pi(a)}{\pi_t(a)} + 1 \right) + \lambda = 0.$$

Consequently,

$$\boxed{\pi_{t+1}(a) = \frac{\pi_t(a) \exp(\eta A_t(a))}{\sum_b \pi_t(b) \exp(\eta A_t(b))}}.$$

Actions outside the support of π_t remain assigned zero probability. The objective is strictly concave on this support, so the optimiser is unique, assuming the normalising constant is finite.
For small η, expand each exponential as $\exp(\eta A_t(a)) = 1 + \eta A_t(a) + O(\eta^2)$. The normalising denominator then becomes

$$\sum_b \pi_t(b) \exp(\eta A_t(b)) = \sum_b \pi_t(b) \big(1 + \eta A_t(b)\big) + O(\eta^2) = 1 + \eta\, \mathbb{E}_{\pi_t}[A_t] + O(\eta^2),$$

using $\sum_b \pi_t(b) = 1$ and $\mathbb{E}_{\pi_t}[A_t] = \sum_b \pi_t(b) A_t(b)$. Applying $\frac{1}{1+x} = 1 - x + O(x^2)$ to this denominator,

$$\pi_{t+1}(a) = \frac{\pi_t(a)\big(1 + \eta A_t(a)\big)}{1 + \eta\, \mathbb{E}_{\pi_t}[A_t]} + O(\eta^2) = \pi_t(a)\big(1 + \eta A_t(a)\big)\big(1 - \eta\, \mathbb{E}_{\pi_t}[A_t]\big) + O(\eta^2).$$

Multiplying out and discarding the $O(\eta^2)$ cross term,

$$\pi_{t+1}(a) = \pi_t(a) \left[1 + \eta \left(A_t(a) - \mathbb{E}_{\pi_t}[A_t]\right)\right] + O(\eta^2),$$

so η controls the update size. Large η places increasingly more mass on high-advantage actions.

(ii) For the true advantage of π_t,

$$J(\pi_t) = \mathbb{E}_{a \sim \pi_t}[A^{\pi_t}(a)] = 0.$$

The regularised objective evaluated at π_t is also zero. Since π_{t+1} maximises it,

$$J(\pi_{t+1}) - \frac{1}{\eta} \mathrm{KL}(\pi_{t+1} \| \pi_t) \geq 0.$$

It follows that

$$\boxed{J(\pi_{t+1}) \geq \frac{1}{\eta} \mathrm{KL}(\pi_{t+1} \| \pi_t) \geq 0 = J(\pi_t).}$$

Thus the update guarantees monotone improvement in the scalar performance measure defined in the question.
In neural-network GRPO the guarantee may fail because the exact advantage is replaced by a noisy finite-sample estimate and the maximisation is replaced by a finite number of stochastic-gradient steps in a restricted parametric policy class. The clipped GRPO objective is also not identical to the exact KL-regularised objective above.

(iii) For one sample, write

$$\ell(\rho, \hat{A}) = \min\left(\rho\hat{A}, \mathrm{clip}(\rho, 1-\epsilon, 1+\epsilon)\hat{A}\right).$$

Equivalently,

$$\ell(\rho, \hat{A}) = \begin{cases} \hat{A}\min(\rho, 1+\epsilon), & \hat{A} \geq 0, \\ \hat{A}\max(\rho, 1-\epsilon), & \hat{A} < 0. \end{cases}$$

Thus, for a positive advantage, the objective gives no further incentive to increase the ratio above $1+\epsilon$. For a negative advantage, it gives no further incentive to reduce the ratio below $1-\epsilon$.
This is not a hard constraint

$$\rho \in [1-\epsilon, 1+\epsilon].$$

Other samples can still move the ratio outside this interval. In contrast, a constraint

$$\mathrm{KL}\big(\pi_\theta(\cdot \mid q)\|\pi_{\mathrm{old}}(\cdot \mid q)\big) \leq \delta$$

is a constraint on the whole distribution $\pi_\theta(\cdot \mid q)$. Although this is a single distribution over the action, the divergence $\sum_a \pi_\theta(a \mid q)\log\frac{\pi_\theta(a|q)}{\pi_{\mathrm{old}}(a|q)}$ depends on the probabilities of all actions simultaneously, including unsampled ones. PPO clipping, by contrast, acts only on the ratio at each individual sampled action and is advantage-dependent; it does not directly constrain unsampled actions, and does not guarantee that this distribution-level KL divergence is small.

I.4.3 Mixture proposal and importance weighting

We suppress the fixed prompt q from the notation. Define

$$p(a) := \pi_{\mathrm{old}}(a \mid q), \qquad u(a) := \pi_{\mathrm{unif}}(a \mid q), \qquad m_\alpha(a) := \pi_{\mathrm{mix}}(a \mid q) = (1-\alpha)p(a) + \alpha u(a),$$

and

$$s_\theta(a) := \nabla_\theta \log \pi_\theta(a \mid q).$$

Assume $m_\alpha(a) > 0$ whenever $p(a) > 0$.

(i) For any integrable function F,

$$\mathbb{E}_{a\sim p}[F(a)] = \mathbb{E}_{a\sim m_\alpha}\left[\frac{p(a)}{m_\alpha(a)}F(a)\right].$$

Define the importance weight

$$w(a) := \frac{p(a)}{m_\alpha(a)}.$$

For a sample $a_i \sim m_\alpha$, this gives

$$w_i := w(a_i) = \frac{\pi_{\mathrm{old}}(a_i \mid q)}{(1-\alpha)\pi_{\mathrm{old}}(a_i \mid q) + \alpha\pi_{\mathrm{unif}}(a_i \mid q)}.$$

Equivalently, when $p(a_i) > 0$,

$$w_i = \left[(1-\alpha) + \alpha\frac{\pi_{\mathrm{unif}}(a_i \mid q)}{\pi_{\mathrm{old}}(a_i \mid q)}\right]^{-1}.$$

Thus an importance-corrected estimator has the form

$$\boxed{\widehat{g}_{\mathrm{mix}} = \frac{1}{K}\sum_{i=1}^{K} w_i\, s_\theta(a_i)\, \widehat{A}_i, \qquad a_i \overset{\mathrm{iid}}{\sim} \pi_{\mathrm{mix}}.}$$

Indeed, for a fixed advantage function A,

$$\begin{aligned}\mathbb{E}_{m_\alpha}\left[w(a)s_\theta(a)A(a)\right] &= \sum_a m_\alpha(a)\frac{p(a)}{m_\alpha(a)}s_\theta(a)A(a)\\ &= \mathbb{E}_p\left[s_\theta(a)A(a)\right].\end{aligned}$$

The importance weight corrects the sampling distribution from m_α back to p. The correct empirical advantage is discussed in part (iii).

(ii) For every fixed realised action a_i,

$$m_\alpha(a_i) \to p(a_i), \qquad w_i = \frac{p(a_i)}{m_\alpha(a_i)} \to 1.$$

Therefore

$$\widehat{g}_{\text{mix}} \to \frac{1}{K}\sum_{i=1}^{K} s_\theta(a_i)\widehat{A}_i.$$

With the standard GRPO empirical advantage

$$\widehat{A}_i = \frac{r_i - \bar{r}}{\sigma_r},$$

we obtain

$$\boxed{\widehat{g}_{\text{mix}} \to \widehat{g}_{\text{GRPO}}.}$$

The same statement holds if a weighted mean and weighted standard deviation are used, since $w_i \to 1$ implies that the weighted empirical statistics converge to the ordinary ones for the same realised actions.

(iii-a) Let

$$\mu_p := V^{\pi_{\text{old}}}(q), \qquad \mu_u := V^{\pi_{\text{unif}}}(q), \qquad \mu_m := V^{\pi_{\text{mix}}}(q).$$

Then, by linearity of expectation,

$$\begin{aligned}V^{\pi_{\text{mix}}}(q) &= \sum_a \big((1-\alpha)p(a) + \alpha u(a)\big)R(q,a)\\ &= (1-\alpha)V^{\pi_{\text{old}}}(q) + \alpha V^{\pi_{\text{unif}}}(q).\end{aligned}$$

Hence

$$\boxed{V^{\pi_{\text{mix}}}(q) = (1-\alpha)V^{\pi_{\text{old}}}(q) + \alpha V^{\pi_{\text{unif}}}(q).}$$

(iii-b) Under mixture sampling,

$$\bar{r} = \frac{1}{K}\sum_{j=1}^{K} r_j \to \mu_m = V^{\pi_{\text{mix}}}(q),$$

not $\mu_p = V^{\pi_{\text{old}}}(q)$. Therefore the unweighted group mean estimates the wrong baseline.
Define

$$A_p(a) := R(q,a) - \mu_p, \qquad A_m(a) := R(q,a) - \mu_m.$$

Then

$$A_m(a) = A_p(a) + \mu_p - \mu_m,$$

where

$$\mu_p - \mu_m = \alpha(\mu_p - \mu_u).$$

Using only the outer importance weight gives

$$\begin{aligned}\mathbb{E}_m[w(a)s_\theta(a)A_m(a)] &= \mathbb{E}_p[s_\theta(a)A_m(a)] \\ &= \mathbb{E}_p[s_\theta(a)A_p(a)] + (\mu_p - \mu_m)\mathbb{E}_p[s_\theta(a)].\end{aligned}$$

Thus

$$\boxed{\mathbb{E}_m[ws_\theta A_m] - \mathbb{E}_p[s_\theta A_p] = (\mu_p - \mu_m)\mathbb{E}_p[s_\theta].}$$

The importance weights correct the outer sampling distribution, but they do not make the unweighted statistic $\bar{r}$ estimate the old-policy value.

At $\theta = \theta_{\text{old}}$, one has the score-function identity

$$\mathbb{E}_p[s_{\theta_{\text{old}}}(a)] = \sum_a p(a)\nabla_\theta \log p(a) = \sum_a \nabla_\theta p(a) = \nabla_\theta 1 = 0.$$

Therefore, for the exact unclipped policy gradient evaluated exactly at the old policy, any action-independent baseline gives the same population gradient. However, the baseline shift changes the empirical advantages and can matter under empirical normalization, PPO clipping, and multiple optimisation steps.

Since

$$\mu_p = \mathbb{E}_m[w(a)R(q,a)],$$

an unbiased raw importance-sampling estimator of the old-policy value is

$$\boxed{\widehat{\mu}_p^{\text{IS}} = \frac{1}{K}\sum_{j=1}^K w_j r_j.}$$

Indeed,

$$\mathbb{E}[\widehat{\mu}_p^{\text{IS}}] = \mathbb{E}_m[wR] = \mu_p.$$

A commonly used practical alternative is the self-normalized estimator

$$\boxed{\widehat{\mu}_p^{\text{SN}} = \frac{\sum_{j=1}^K w_j r_j}{\sum_{j=1}^K w_j}.}$$

This is consistent because

$$\mathbb{E}_m[w] = 1,$$

but it is generally slightly biased for finite K.

A corresponding self-normalized weighted empirical variance is

$$\boxed{\left(\widehat{\sigma}_p^{\text{SN}}\right)^2 = \frac{\sum_{j=1}^K w_j (r_j - \widehat{\mu}_p^{\text{SN}})^2}{\sum_{j=1}^K w_j}.}$$

Thus a corrected estimator can be written as

$$\boxed{\widehat{g}_{\text{mix}}^{\text{corr}} = \frac{1}{K}\sum_{i=1}^K w_i s_\theta(a_i)\frac{r_i - \widehat{\mu}_p}{\widehat{\sigma}_p}.}$$

Equivalently, absorb the importance weight into the advantage:

$$\boxed{\widetilde{A}_i = w_i \frac{r_i - \widehat{\mu}_p}{\widehat{\sigma}_p}, \qquad \widehat{g}_{\text{mix}}^{\text{corr}} = \frac{1}{K}\sum_{i=1}^K s_\theta(a_i)\widetilde{A}_i.}$$

Here $\widehat{\mu}_p$ may be either the raw estimator $\widehat{\mu}_p^{\mathrm{IS}}$ or the self-normalized estimator $\widehat{\mu}_p^{\mathrm{SN}}$. The raw estimator is unbiased for μ_p; the self-normalized estimator is often more stable but is generally not exactly finite-sample unbiased.

For an exactly unbiased finite-sample gradient without empirical standardization, one may use the leave-one-out baseline

$$\widehat{\mu}_{p,-i}^{\mathrm{IS}} = \frac{1}{K-1}\sum_{j\neq i} w_j r_j,$$

giving

$$\boxed{\widehat{g}_{\mathrm{LOO}} = \frac{1}{K}\sum_{i=1}^{K} w_i s_\theta(a_i)\left(r_i - \widehat{\mu}_{p,-i}^{\mathrm{IS}}\right).}$$

Because $\widehat{\mu}_{p,-i}^{\mathrm{IS}}$ is independent of a_i and has expectation μ_p, this is unbiased for

$$\mathbb{E}_p[s_\theta(a)(R(q,a)-\mu_p)].$$

Empirical standardization by a random $\widehat{\sigma}_p$ is generally a variance-control heuristic rather than an exactly unbiased operation.

(iv) Let

$$A_p(a) := R(q,a)-\mu_p, \qquad Z(a) := s_\theta(a)A_p(a), \qquad g := \mathbb{E}_p[Z(a)].$$

Ignoring empirical baseline and variance estimation, the old-policy estimator has covariance

$$\mathrm{Cov}(\widehat{g}_{\mathrm{GRPO}}) = \frac{1}{K}\left(\mathbb{E}_p[ZZ^\top] - gg^\top\right).$$

The importance-sampled estimator has covariance

$$\begin{aligned}\mathrm{Cov}(\widehat{g}_{\mathrm{mix}}) &= \frac{1}{K}\left(\mathbb{E}_m[w^2 ZZ^\top] - gg^\top\right)\\ &= \frac{1}{K}\left(\mathbb{E}_p[wZZ^\top] - gg^\top\right).\end{aligned}$$

Therefore

$$\boxed{\mathrm{Cov}(\widehat{g}_{\mathrm{mix}}) - \mathrm{Cov}(\widehat{g}_{\mathrm{GRPO}}) = \frac{1}{K}\mathbb{E}_p[(w-1)ZZ^\top].}$$

There is no universal ordering: importance sampling can either increase or decrease variance.

For the scalar advantage contribution alone,

$$\boxed{\mathrm{Var}_m(wA_p) - \mathrm{Var}_p(A_p) = \mathbb{E}_p[(w-1)A_p^2].}$$

For $\alpha > 0$,

$$w(a) > 1 \iff m_\alpha(a) < p(a) \iff u(a) < p(a),$$

and

$$w(a) < 1 \iff m_\alpha(a) > p(a) \iff u(a) > p(a).$$

Therefore, if large-magnitude advantages occur mainly on actions satisfying $u(a) < p(a)$, then $w(a) > 1$ where $A_p(a)^2$ is large, so

$$\mathbb{E}_p[(w-1)A_p^2] > 0,$$

and the variance increases. Intuitively, the mixture samples these old-policy-common actions less frequently and compensates with weights larger than one.

Conversely, if large-magnitude advantages occur mainly on actions satisfying $u(a) > p(a)$, then $w(a) < 1$ where $A_p(a)^2$ is large, so

$$\mathbb{E}_p[(w-1)A_p^2] < 0,$$

and the variance decreases. Intuitively, the uniform component oversamples old-policy-rare but high-magnitude-advantage responses, reducing their importance weights.

A complementary calculation for the unweighted reward variability is the mixture-variance identity. Let

$$\sigma_p^2 := \mathrm{Var}_p(R), \qquad \sigma_u^2 := \mathrm{Var}_u(R).$$

Then

$$\boxed{\mathrm{Var}_{m_\alpha}(R) = (1-\alpha)\sigma_p^2 + \alpha\sigma_u^2 + \alpha(1-\alpha)(\mu_p - \mu_u)^2.}$$

Therefore

$$\mathrm{Var}_{m_\alpha}(R) - \sigma_p^2 = \alpha\left[\sigma_u^2 - \sigma_p^2 + (1-\alpha)(\mu_p - \mu_u)^2\right].$$

Thus the raw reward variance increases if

$$\sigma_u^2 + (1-\alpha)(\mu_p - \mu_u)^2 > \sigma_p^2,$$

for example if the uniform policy has much larger reward variance. It decreases if

$$\sigma_u^2 + (1-\alpha)(\mu_p - \mu_u)^2 < \sigma_p^2,$$

for example if $\mu_u = \mu_p$ and $\sigma_u^2 < \sigma_p^2$.

(v-a) Since $0 < c < 1$ we have $\log c < 0$, so

$$\boxed{w_i = c^T = e^{T \log c} \longrightarrow 0 \ \text{ exponentially fast as } T \to \infty.}$$

A value $c < 1$ is the typical situation when $\alpha > 0$ because the responses are drawn from π_{mix}, whose uniform component inflates the probability of tokens that are unlikely under π_{old}, so the sampled tokens usually satisfy $\pi_{\text{mix}} > \pi_{\text{old}}$, i.e. $c = \pi_{\text{old}}/\pi_{\text{mix}} < 1$.

(v-b) When $w_i = c^T$ is exponentially small, almost every sampled trajectory contributes negligible gradient signal, so $\widehat{g}_{\text{mix}}$ is dominated by a handful of rare large-weight trajectories: the effective sample size collapses and the variance explodes, making the estimate unreliable for long responses.

One concrete fix is to shrink the mixture weight with length, taking $\alpha = \alpha_T = O(1/T)$ so that the per-token ratio satisfies $c_T = 1 - \kappa/T + o(1/T)$ and the full-response weight stays bounded away from zero,

$$w_i = c_T^T \longrightarrow e^{-\kappa} > 0.$$

The trade-off is weaker uniform exploration on long responses.